

guardrails-cpp: Sub-Millisecond Native C++20 and CUDA Security Guardrail Engine for Production LLM Serving

Kamran Saberifard

Department of Computer Engineering, MehrAsthan University, Guilan, Iran
Aryorithm Group, Global Systems & AI Security Research Division
kamisaberi@gmail.com — ORCID: [0009-0002-7822-6168](https://orcid.org/0009-0002-7822-6168)

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Abstract

Large Language Models (LLMs) deployed in production enterprise environments require real-time security guardrails to defend against prompt injection attacks, jailbreak attempts, toxic content generation, and Personally Identifiable Information (PII) exfiltration. However, existing standard guardrail frameworks—such as Python-based **Guardrails AI** and NVIDIA **NeMo Guardrails**—introduce prohibitive latency penalties ranging from 50ms to over 200ms per request. This overhead is caused by interpreted string parsing, Global Interpreter Lock (GIL) contention, redundant buffer allocations, and un-optimized model wrapper calls. In this paper, we present **guardrails-cpp**, a native C++20 and CUDA security engine engineered specifically for high-throughput LLM serving runtimes (e.g., vLLM, TensorRT-LLM). **guardrails-cpp** replaces Python-level validation pipelines with zero-copy C++ memory management, SIMD-accelerated Aho-Corasick prompt injection pattern matching, zero-allocation PII redaction, embedded C++ LibTorch/TensorRT safety classifier inference, and custom CUDA token stream scanning kernels. Benchmark evaluations on an NVIDIA RTX 4090 GPU demonstrate that **guardrails-cpp** executes the complete multi-layer security pipeline in **0.45ms** per request—achieving a **175.7× throughput improvement** over Python baselines while enabling sub-millisecond real-time threat mitigation. The system is open-sourced to advance secure AI serving infrastructure.

Keywords: LLM Security, Prompt Injection, Jailbreak Defense, CUDA Kernel, C++ LibTorch, High-Performance Computing, AI Guardrails, Real-Time Inference.

1 Introduction

Large Language Models (LLMs) have become core infrastructure in modern software ecosystems, powering autonomous agents, customer service systems, and code generation platforms [1]. However, deploying LLMs introduces critical security vulnerabilities. Adversaries can exploit prompt injections [2], jailbreak sequences (e.g., "Do Anything Now" / DAN prompts) [3], and data exfiltration vectors to force models into violating system policies, revealing confidential instructions, or leaking Personally Identifiable Information (PII).

To mitigate these risks, organizations deploy "guardrails"—validation layers that inspect incoming user prompts and outgoing generated text against safety policies. Popular open-source solutions such as **Guardrails AI** [4] and **NeMo Guardrails** [5] provide policy abstractions written in Python.

Despite their functional utility, existing Python-based guardrail architectures introduce severe system-level performance bottlenecks:

1. **Prohibitive Processing Latency:** Python string manipulations and regex operations add 50ms–200ms of latency per query, often exceeding the execution time of the LLM inference itself.
2. **Interpreter GIL Bottlenecks:** Multithreaded Python API gateways suffer from Global Interpreter Lock (GIL) contention when handling high-concurrency token streams.
3. **Un-Optimized Memory Buffering:** Passing prompt text between Python string objects, regex engines, and PyTorch model wrappers causes excessive heap allocations and CPU-to-GPU memory copies.

To resolve these latency barriers, we introduce **guardrails-cpp**, a native C++20 and CUDA guardrail engine designed for seamless proxy integration with high-performance LLM serving systems like vLLM [6] and TensorRT-LLM.

2 Threat Model & Defense Architecture

guardrails-cpp enforces a strict four-stage security pipeline prior to feeding user prompts into the LLM tokenization and GPU Key-Value (KV) cache memory spaces.

2.1 1. Stage 1: SIMD Prompt Injection Jailbreak Scanner

Adversaries frequently use adversarial prompt prefixes (e.g., *"Ignore all previous instructions"*) to override model system prompts. **guardrails-cpp** utilizes a multi-pattern search algorithm optimized with SIMD instruction sets to scan prompt text at gigabytes-per-second memory bandwidth speeds.

2.2 2. Stage 2: Zero-Copy SIMD PII Sanitizer

Before prompts reach model context windows, sensitive user identifiers—including Social Security Numbers (SSNs), Credit Card Numbers, API Secret Keys, and Email addresses—are identified using C++ regular expression engines and redacted in-place with zero memory allocation.

2.3 3. Stage 3: Embedded C++ Safety Classifier

For semantic toxicity and policy classification, **guardrails-cpp** embeds Meta’s Llama-Guard [7] safety model directly within a native C++ LibTorch/TensorRT engine, executing forward inference passes without leaving C++ process space.

2.4 4. Stage 4: Real-Time CUDA GPU Token Stream Scanner

During streaming output generation, an outgoing token stream can violate safety policies mid-generation. **guardrails-cpp** executes a custom CUDA kernel (`scan_token_stream_kernel`) that inspects active GPU token memory buffers in real-time, instantly setting atomic cancellation flags if banned tokens appear.

3 CUDA Kernel Engineering

Listing 1 presents the custom CUDA kernel implementation designed to scan GPU token streams during streaming LLM text generation.

```

1 __global__ void scan_token_stream_kernel(
2     const int32_t* __restrict__ tokens,
3     int num_tokens,
4     const int32_t* __restrict__ banned_tokens,
5     int num_banned,
6     int32_t* __restrict__ violation_flag)
7 {
8     int tid = blockIdx.x * blockDim.x + threadIdx.x;
9
10    if (tid < num_tokens) {
11        int32_t token = tokens[tid];
12
13        // Compare generated GPU token against array of banned token IDs
14        for (int b = 0; b < num_banned; ++b) {
15            if (token == banned_tokens[b]) {
16                // Atomic flag update to halt streaming generation on host
17                atomicExch(violation_flag, 1);
18                break;
19            }
20        }
21    }
22 }

```

Listing 1: Fused CUDA Kernel for Real-Time GPU Token Stream Safety Filtering.

4 Experimental Evaluation

4.1 Benchmark Setup

We evaluated `guardrails-cpp` against Python Guardrails AI (v0.4.0) and NeMo Guardrails (v0.8.0).

- **Hardware:** NVIDIA RTX 4090 GPU (24 GB VRAM), AMD Ryzen 9 7950X 16-Core Processor, 64 GB DDR5 RAM.
- **Software:** Ubuntu 22.04 LTS, CUDA 12.2, LibTorch 2.1.2 C++ API, GCC 11.4.0.
- **Dataset:** 1,000 synthetic evaluation prompts (average length 512 tokens) containing mixed prompt injections, PII entities, and benign queries.

4.2 Latency Throughput Evaluation

Table 1 provides a stage-by-stage latency breakdown comparing Python guardrails against `guardrails-cpp`.

Table 1: Stage-by-stage latency (ms) and overall throughput (req/sec) comparison on NVIDIA RTX 4090.

Pipeline Stage	Guardrails AI (Python)	NeMo Guardrails (Python)	<code>guardrails-cpp</code>
Prompt Injection Check	18.40 ms	24.10 ms	0.12 ms
PII Masking & Redaction	12.20 ms	15.80 ms	0.08 ms
Safety Classifier Pass	48.50 ms	52.30 ms	0.25 ms
Total Latency	79.10 ms	92.20 ms	0.45 ms
Throughput (req/sec)	12.6 req/sec	10.8 req/sec	2,222.2 req/sec

As shown in Table 1, `guardrails-cpp` reduces overall validation latency from **79.10ms** down to **0.45ms**, delivering a **175.7 $\times$ speedup** and achieving a throughput of over **2,200** requests per second.

5 Conclusion

In this paper, we introduced `guardrails-cpp`, a native C++20 and CUDA security engine for LLM serving systems. By executing prompt injection checks, PII redactions, Llama-Guard inferences, and GPU token stream scans entirely within native C++ and custom CUDA kernels, `guardrails-cpp` reduces guardrail validation latency to under **0.5ms**, removing security as a performance bottleneck in enterprise AI serving.

Open Source Availability

The complete source code, CMake build system, and benchmarks are available at: <https://github.com/kamisaberi/guardrails-cpp>.

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